HARD · PROBLEM-SOLVING AND DATA ANALYSIS

Mixed

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

An area of 46.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k? $1 \quad \text{nautical} \quad \text{mile} = 1.852 \quad \text{kilometers}$

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

Sample	Percent in favor	Margin of error
A	52%	4.2%
B	48%	1.6%

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

A. Sample A had a smaller number of votes that could not be recorded.

B. Sample A had a higher percent of favorable responses.

C. Sample A had a larger sample size.

D. Sample A had a smaller sample size.

Q3

GRID-IN ALGEBRA

$$6 + 7r = pw$$

$$7r - 5w = 5w + 11$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN ALGEBRA

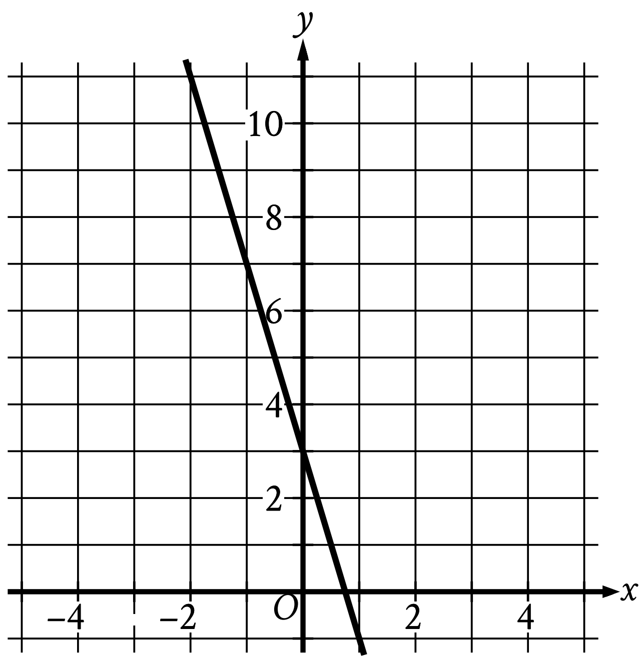
$$I = \frac{V}{R}$$

The formula above is Ohm's law for an electric circuit with current I, in amperes, potential difference V, in volts, and resistance R, in ohms. A circuit has a resistance of 500 ohms, and its potential difference will be generated by n six-volt batteries that produce a total potential difference of $6n$ volts. If the circuit is to have a current of no more than 0.25 ampere, what is the greatest number, n, of six-volt batteries that can be used?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (negative 1 comma 7)
  - (0 comma 3)

The graph of the linear function $y = fx + 19$ is shown. If c and d are positive constants, which equation could define f?

A. $fx =$ -d - cx

B. $fx = d$ - cx

C. $fx =$ -$d + cx$

D. $fx = d + cx$

Q6 GRID-IN GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has a diameter with endpoints $2, 4$ and $2, 14$. An equation of this circle is $x - 2^2 + y - 9^2 = r^2$, where r is a positive constant. What is the value of r?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7 GEOMETRY AND TRIGONOMETRY

Circle A has equation $x - 7^2 + y + 3^2 = 1$. In the xy-plane, circle B is obtained by translating circle A to the right 4 units. Which equation represents circle B?

A. $x - 7^2 + y + 7^2 = 1$

B. $x - 3^2 + y + 3^2 = 1$

C. $x - 11^2 + y + 3^2 = 1$

D. $x - 7^2 + y - 1^2 = 1$

Q8

GRID-IN ADVANCED MATH

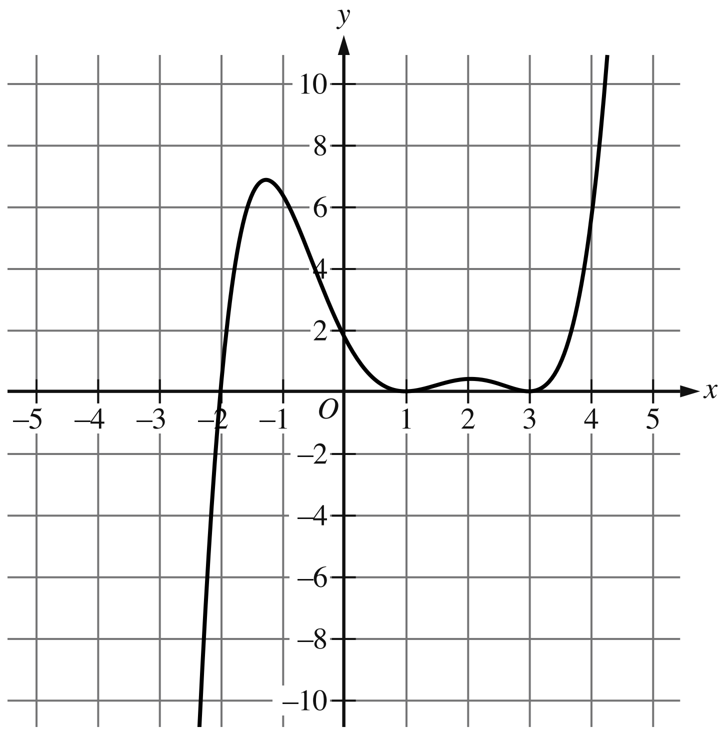
$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8 - \sqrt{k}}{4}$, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right, the curve passes from quadrant 3 to quadrant 2 to quadrant 1.
- The curve has 2 relative maximums and 2 relative minimums.
- The curve crosses the x axis at the point (negative 2 comma 0).
- The curve touches the x axis at the following points:
  - (1 comma 0)
  - (3 comma 0)

The graph of $y = gx$ is shown. If $gx + 28 = \frac{1}{10}x - a^2x - b^2x - c$, where a, b, and c are constants, which of the following could be the value of b?

A. 31

B. 3

C. -2

D. -25

Q10 GRID-IN ADVANCED MATH

$$x - 9 + 45 = 63$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.